WHITEBOX: Which Explanations of a Neural Network Are Real?

An Overview for a Mathematician

Arjun Paradkar
Department of Mathematics, University of Arkansas
paradkar@uark.edu

Abstract. Modern neural networks are called "black boxes" because their internal computation cannot be read. A research field exists to look inside them, but it rests on an unexamined assumption: that what it finds is a property of the network rather than of the coordinates in which the network happens to be written. This document explains the problem in plain terms, reviews seven specific things people mean by "black box" and how far each has been solved, describes honestly what went wrong in three earlier versions of this project, and sets out the current program: a symmetry-based test of which internal measurements are meaningful, a spectral test of what current tools miss, and a structural test of which systems can be explained at all. It closes with three open mathematical questions — on the completeness of a symmetry group, the generators of an invariant ring, and fixed-parameter tractability in a graph-width parameter — that a topologist is well placed to attack.

1 The problem in plain terms

Modern artificial intelligence systems are built by a procedure that nobody fully understands. You take a very large function with billions of adjustable numbers (the "parameters"), show it enormous amounts of data, and nudge the numbers until the function does what you want. It works astonishingly well. But when you open the finished system up and look at the numbers, you cannot read them. There is no place where "the concept of a cat" or "the rule for addition" lives. The knowledge is spread across everything in a way we cannot decode.

This is what people mean when they call these systems a "black box." It matters because the systems are now used for medical decisions, hiring, and safety-critical tasks, and nobody can verify what they actually learned.

A whole research field — called interpretability — tries to look inside. It has produced many tools and many findings. But it has a problem that sits underneath all of them, and that problem is what this project is about: **when a researcher finds something inside a network, nobody can check whether the finding is a real property of the network or an artifact of how the numbers happened to be written down.**

2 The seven original problems, and how much each is solved

When we started, we listed seven distinct things people mean by "black box." Here is each one, what the field has done about it, what our first version claimed, and what honestly survived.

Problem	What the field has	Honest status
1. What did it learn?	Methods that find "features" (directions in the network's internal space that respond to concepts). Real but incomplete, and never checked for whether they depend on coordinates.	Partially solved by others. Our first version's contributions here did not replicate.
2. Why did it make this decision?	Well-developed methods that trace a single output back to its inputs.	Mostly solved by others. We never claimed this one.
3. When did bias enter during training?	Tools that trace behavior back to training data. Nothing that watches bias form structurally.	Open. Our first version claimed an early-warning signal; it failed to replicate at every scale we tested.
4. What does it not know?	Uncertainty estimates on outputs. Nothing structural.	Open on the model side. We can now measure what our *tools* miss (see below).
5. Where will it fail before it fails?	Robustness testing. Nothing predictive from internal structure.	Open. Our attempt was a clean null result.
6. Is it memorizing or learning?	Several methods that detect memorization without a test set, cheaper than ours.	Partially solved by others. Our headline number was an artifact; one narrow, honest signal survived.
7. How do its internal representations change during training?	An active research area with real theorems.	Partially solved by others. Our version had one small genuine sliver.

The honest tally: none of the seven is fully solved by anyone. Problem 2 is closest. We solved zero of them in our earlier versions. What we own is one narrow validated signal and — more importantly — the discipline of controls that told us the rest was wrong.

3 What went wrong in our earlier versions

Three versions, three different mistakes. I am including this because the mistakes are what taught us what the real problem is.

First version (the competition paper). We used a topological summary of the network's internal states: build a point cloud from the activations, compute persistent homology, and sum the squared lifetimes of the features. We found this number was about three thousand times larger in "learning" networks than in "memorizing" ones and called it a breakthrough. The error: that number is dominated by how spread out the cloud is, not by its shape. Rescale the cloud by a factor λ and the number scales by λ^2. Our "memorizing" networks had never learned anything at all; their internal states had simply shrunk. We were measuring variance and calling it topology. A simple control — shuffle the data so all real structure is destroyed while keeping each coordinate's distribution — retained most of the number. That should have been run first. It was not.

Second version (the software library with proper controls). We rebuilt everything with the controls in place, and the controls destroyed most of the first version's claims. One signal survived, but only when calibrated separately for each type of network, because the effect actually reverses sign between image networks and language networks. We also tested a hypothesis that a network which truly understands its data should have internal structure matching the shape of the data. The opposite is true: a network that classifies well throws away the input's shape, because it does not need it.

Third version (two weeks ago). We designed an experiment to test whether the "many competing explanations" problem was really a coordinate artifact — and I built a tautology into it. The details are in

the mathematical companion document. The short version: I asked whether two different sub-circuits of the *same* network were related by a symmetry, and for a generic network the answer is forced to be "no" regardless of anything true about explanations. The controls caught it before it became a claim.

The pattern: every version tried to measure structure before checking whether the measurement itself was meaningful. The current version reverses that order.

4 The three root problems we now attack

We traced the seven symptoms to three underlying causes. Solving the causes moves the symptoms.

4.1 Root 1: nobody can tell which explanations are real

Here is the central idea, and it is a mathematical one. A neural network can be rewritten in many ways that change nothing about what it computes. You can relabel the hidden units in any order. You can multiply the weights going into a unit by a positive constant and divide the weights coming out by the same constant. These rewritings form a group — the network's symmetry group. Two parameter settings related by a symmetry are the same computation in different coordinates, exactly as two coordinate charts describe one manifold.

Now: any quantity someone measures inside a network that *changes* when you apply a symmetry was never a property of the network. It was a property of the coordinates. This is a criterion the field has accepted since 2017 for one narrow class of methods (attributing a decision to its inputs). Nobody has ever applied it to the network's internals — the circuits, the features, the "most important neurons." We are running that audit. Our expectation is that several widely used measurements will fail it.

The same idea answers a second question: is the mechanism behind a behavior unique? If two independently trained networks solve a problem in ways that are the same up to symmetry, there is one mechanism. If not, there are genuinely several, and now they can be counted. We have already learned something here: when you look *within* one network, the multiple valid circuits are not related by symmetry at all — they are structural redundancy, combinatorially many ways of recombining duplicated parts. So the field's "many explanations" problem has two distinct sources, and only one of them is coordinates. That separation is new.

4.2 Root 2: nobody has measured what the tools miss

The field's best tool for finding features inside a network is a kind of dictionary-learning method. It reconstructs the network's internal states approximately. The approximation error is large — replacing a leading system's internal states with the reconstruction costs it capability equivalent to roughly ninety percent of its training effort. Nobody has properly tested whether that leftover error is noise or is itself doing computation. There is a standard mathematical tool for exactly this question: the eigenvalue distribution of a pure-noise covariance matrix is known exactly (the Marchenko–Pastur law), so any eigenvalue that escapes the predicted bulk is structure. We apply it. If the leftover is structured, the field's decompositions are blind to working machinery, and we will have quantified how much.

4.3 Root 3: nobody knows which systems can be explained at all

It has been proven that finding the circuit responsible for a behavior is computationally intractable in general — a hardness result of the kind that does not go away with faster computers. Yet in practice it sometimes works. What separates the networks where it works from the ones where it does not? Our hypothesis is that it is structural: networks whose internal connectivity is organized into modules are tractable, and those that are not, are not. We measure this across many networks and look for the transition. If we find it, explainability becomes a property you can measure before you spend the effort — and it raises the deeper question of whether the systems that *work* are the ones that are explainable, which is where this connects to biology.

5 The experiments, by machine

Everything is pre-registered: hypotheses and pass/fail thresholds are written down before any computation runs. Every positive result is attacked with a control designed to destroy it. Negative results are written up with the same care.

5.1 On the laptop (ordinary processor, running now)

- **Small networks with complete ground truth.** Tiny networks trained on Boolean functions, small enough that every valid sub-circuit and every valid interpretation can be enumerated exhaustively. This is where the symmetry questions are answered exactly. About 3,600 such networks have been trained.
- **The symmetry group of each network,** including a check for hidden symmetries beyond the obvious ones, via the rank of a Jacobian.
- **The cross-network uniqueness test:** are the circuits found in independently trained networks the same up to symmetry?
- **The audit table:** every candidate measurement evaluated across hundreds of symmetry-transformed copies of each network, classified as invariant, predictably-varying, or artifact.
- **The software itself:** the auditing function, the null-model library, the topology tools.

5.2 On the local graphics card (a consumer GPU)

- **The audit on a real language model** (a small public one), to show the results are not an artifact of toy networks.
- **The dark-matter test** on publicly released dictionary decompositions: is the leftover noise or machinery?
- **The legibility ensemble:** about sixty small networks with varying structure, measuring how hard each is to explain and what predicts it.
- **Re-examination of our earlier claims** under the new criteria: which survive, which are formally retired.

- **Topology beyond persistent homology:** tracking individual topological features across training time (zigzag persistence), two-parameter persistence, and topological summaries of the network's connectivity graph.

5.3 On the university cluster (seventy GPUs, pending access)

- **The audit on a frontier-class model** (eight to nine billion parameters). This is what turns "true for small networks" into "a statement about real AI."
- **Dark matter at scale,** including whether larger dictionaries ever close the gap.
- **The legibility question on a public model with all its training checkpoints released,** so anyone can verify whether it became more explainable as it learned.
- **Running the whole instrument suite end to end** on a large model, to confirm it is usable.

6 The end goals

6.1 Near: the instruments

A software library any laboratory can run on any model, with four functions:

- *Is this explanation real?* — the symmetry audit.
- *Is there one mechanism or several?* — the uniqueness test, plus a measure of whether local explanations on different inputs can be glued into one global explanation (this is a Čech cohomology computation; the obstruction is H^1).
- *How much did your tool miss?* — the dark-matter measurement.
- *Can this system be explained at all?* — the structural predictor.

Every function ships with the control it was tested against and a written statement of where it fails. The library is deliberately *not* described as "opening the black box." It is the set of instruments that tells you whether anyone's opening — including ours — is real. Given that the field has produced hundreds of "openings" nobody could verify, that may be the more useful thing.

6.2 Far: understanding as a transportable object

Once measurements are certified, the next step is to extract the certified structures from a system that demonstrably understands something (we start with small networks that have provably learned modular arithmetic), rebuild the capability from those structures alone, and show the rebuild behaves identically. If that works, understanding becomes something you can read out of one system and write into another. The eventual biological anchor is the mammalian grid-cell system, where it is already established — by topological methods — that the brain represents position on a torus. It is one of very few places where a brain demonstrably uses an explicit mathematical object as its representation, which makes it the right place to test whether artificial and biological understanding share a structure. The far endpoint is a system built from certified structures from the start: legible by design, because we will finally know what legible means.

7 Where your expertise matters

Three of the questions in this program are, at bottom, theorems waiting to be proved or disproved, and they are in your territory.

1. **Completeness of the symmetry group.** For the class of networks we study, does the group we can write down (relabelings and balanced rescalings) account for the entire fiber of the parameter-to-function map, or are there hidden symmetries? We can detect hidden ones numerically; a proof of when they do and do not exist would settle the foundation.
2. **Generating invariants.** If the meaningful quantities are exactly the functions on parameter space invariant under this group, is there a finite generating set — an analogue of the First Fundamental Theorem of invariant theory for this particular product of permutation and scaling groups? An answer tells us whether our "dictionary" of meaningful measurements can ever be complete.
3. **Tractability from structure.** Circuit discovery is NP-hard in general. Is it fixed-parameter tractable in a modularity parameter of the network's connectivity graph? The empirical phase diagram is evidence; the theorem is the prize.

The companion document sets out the mathematics behind every experiment in full.

Three versions of this project failed by measuring structure before checking whether the measurement was real. This version checks first. Everything above is built so that the fourth version cannot fail the same way.